DIGITAL IMAGE PROCESSING HOMEWORK

HOMEWORK 1

卞常灿 自动化3班
2023312216

2.9 A common measure of transmission for digital data is the *baud rate*, defined as symbols (bits in our case) per second. As a minimum, transmission is accomplished in packets consisting of a start bit, a byte (8 bits) of information, and a stop bit. Using these facts, answer the following:

(a)* How many seconds would it take to transmit a sequence of 500 images of size 1024 × 1024 pixels with 256 intensity levels using a 3 M-baud (10^6 bits/sec) baud modem? (This is a representative medium speed for a DSL (Digital Subscriber Line) residential line.

(b) What would the time be using a 30 G-baud (10^9 bits/sec) modem? (This is a representative medium speed for a commercial line.)

Solution:

(a) $N \times M \times L \times 500 \div 10^6$

$= 1024 \times 1024 \times 256 \times 500 \div 10^6 \approx 134218\,s$

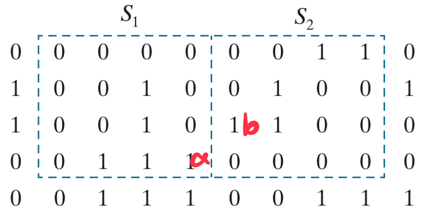
(b) $1024 \times 1024 \times 256 \times 500 \div 10^9 \approx 135\,s$

2.14 Consider the two image subsets, S_1 and S_2 in the following figure. With reference to Section 2.5, and assuming that $V = \{1\}$, determine whether these two subsets are:

(a)* 4-adjacent.

(b) 8-adjacent.

(c) *m*-adjacent.



判断两个子集是否邻接：
只要∃p∈S_1，q∈S_2，p与q邻接，
那么就说S_1和S_2是邻接的

V={1}，所以只看标注有1的点。

4-邻接：∀p∈S_1，都没有S_2中的q在它的上下左右，所以不是4-adjacent

8-邻接：如图，a∈S_1，b∈S_2，且b在a的对角线上，所以是8-adjacent

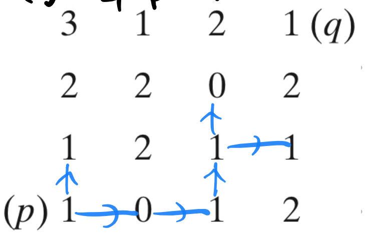
m-邻接：如图a∈S_1，b∈S_2，且b在a的对角线上且$N_4(a) \wedge N_4(b)$中没有1，所以是m-adjacent

2.18 Consider the image segment shown in the figure that follows.

(a)*As in Section 2.5, let $V = \{0,1\}$ be the set of intensity values used to define adjacency. Compute the lengths of the shortest 4-, 8-, and *m*-path between *p* and *q* in the following image. If a particular path does not exist between these two points, explain why.

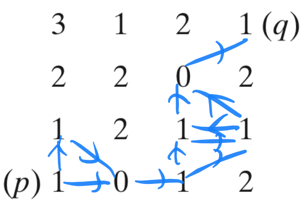
	3	1	2	1 (*q*)
	2	2	0	2
	1	2	1	1
(*p*)	1	0	1	2

(a) 4-path



No 4-path exists, as there is no other pixels with the value of 0 or 1 that is 4-adjanct to the pixel q

8-path shortest length: 5



m-path shortest length: 5